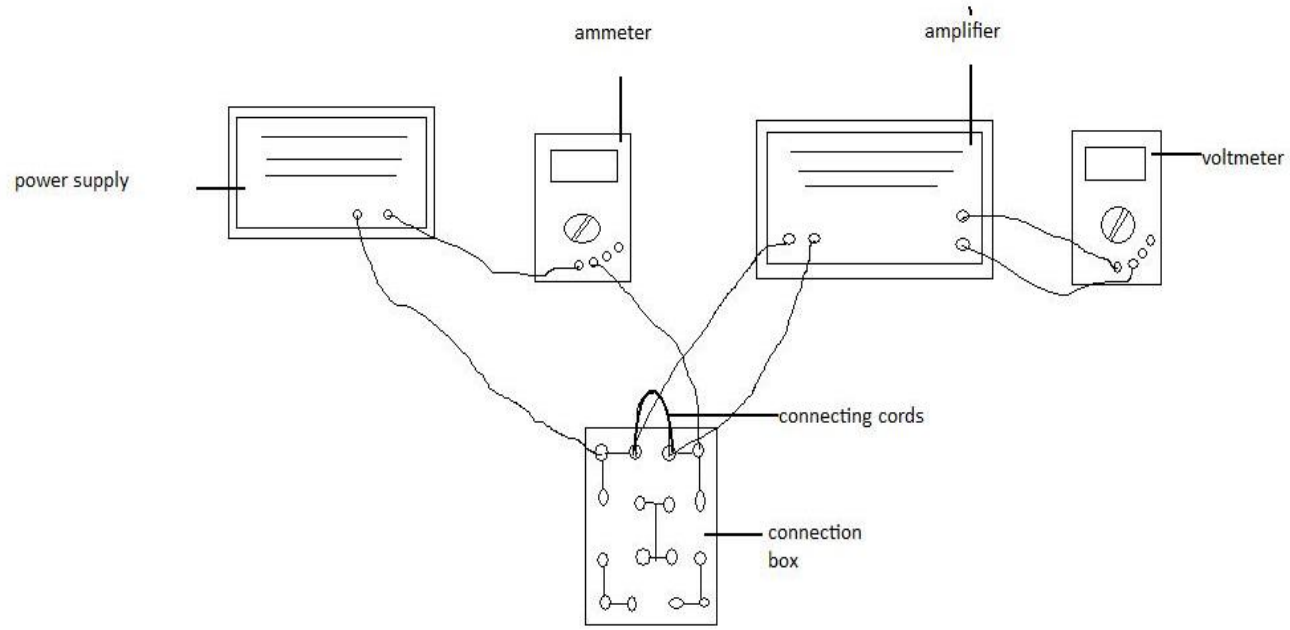
AIM

The aim of the experiment is to determine the resistances of some connecting cords of different lengths by plotting their current/voltage characteristics.

APPARATUS

- Heat conductivity rod, Cu 1
- Heat conductivity rod, Al 1
- Universal measuring amplifier 1
- Digital multimeter 2
- Power supply 0-12V DC/6V, 12VAC 1
- Connection box 1
- Connecting cord, l= 100mm 2
- Connecting cord, l= 250mm 1
- Connecting cord, l= 250mm 1
- Connecting cord, l= 500mm 2
- Connecting cord, l=500 mm 1
- Connecting cord, l= 750mm 2
- Connecting cord, l= 750mm 1
- Connecting cord, l= 2000mm 2
- Connecting cords (different lengths) 3

DIAGRAM



THEORY

Resistivity of a metal conductor can be determined from the resistance R of the metal conductor and its dimensions. The resistance of a metal conductor is the ability of the conductor to inhibit the flow of current passing through it. From Ohm's law which states that when a voltage (potential difference) ΔV is applied across the ends of a conductor, the current in the conductor is found to be proportional to the applied voltage. This law will hold if the temperature of the conductor is kept constant.

The resistance of a piece of wire depends on its length, and its cross-sectional area, or thickness. The longer the wire is, the greater its resistance. If one wire is twice as long as a wire of identical diameter and material, the longer wire offers twice as much resistance as the shorter one. A thicker wire, however, has less resistance, because a thick wire offers more room for an electric current to pass through than a thin wire does. Scientists describe this relationship between resistance, length, and area by saying that resistance is proportional to length and inversely proportional to cross-sectional area. That is:

$$R \propto \frac{l}{A}$$

$$\text{Also} \quad I \propto \Delta V$$

$$\Delta V = IR$$

Where V is the potential difference, *I* is the current and R is the resistance. A graph of ΔV against *I* is a straight line through the origin with a slope equal to the resistance R.

The resistivity of the conductor can therefore be determined since the resistance is directly proportional to the length and inversely proportional to the area.

$$R = \rho \frac{l}{A}$$

Where R is the resistance of the metal rod, ρ is its resistivity, l is the length of the rod and A is the cross-sectional area of the rod.

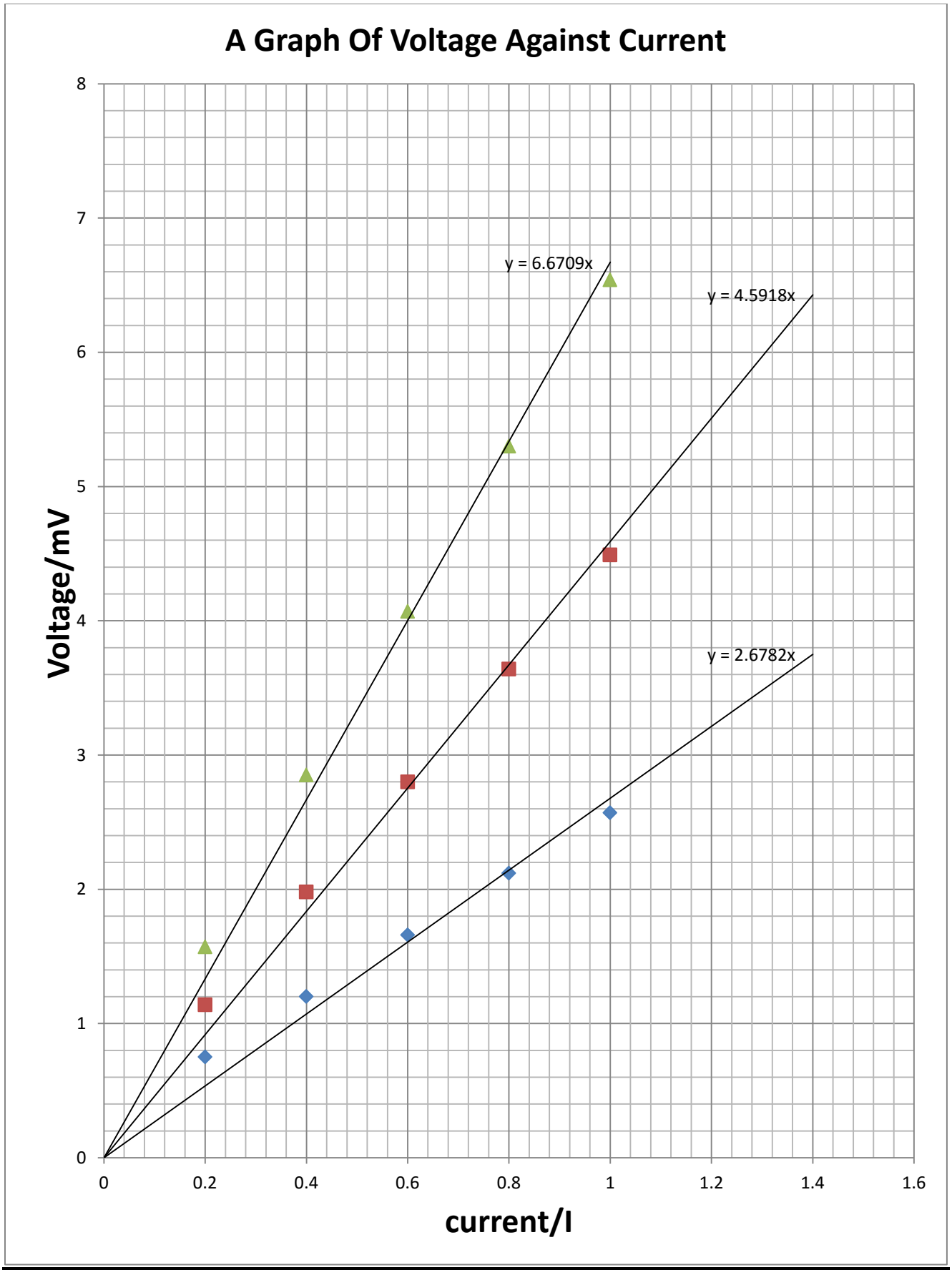
PROCEDURE

The lengths of the connecting cords provided were measured and recorded. Using the connection box, one of the connecting cords provided were connected in series with the powers supply and the ammeter. The input of the amplifier was connected to the sockets of the connecting cord connector and the voltmeter was connected to the output of the amplifier. The voltage source was set to 12V and the current was varied at constant intervals of 0.2A.

The voltage drop across the cord was measured and recorded for each current selected. The experiment was repeated for the two remaining connecting cords provided. On the same graph, voltage V against current I was plotted for each of the cords. The slope for each graph was determined.

TABLE OF RESULTS

Connecting cord/ 36cm		**Connecting cord/ 61cm**		**Connecting/ 86cm**	
Current/ I	**voltage/ mV**	**Current/ I**	**voltage/ mV**	**Current/ I**	**voltage/ mV**
0.20	0.75	0.20	1.14	0.20	1.57
0.40	1.20	0.40	1.98	0.40	2.85
0.60	1.66	0.60	2.80	0.60	4.07
0.80	2.12	0.80	3.64	0.80	5.30
1.00	2.57	1.00	4.49	1.00	6.54


A Graph Of Voltage Against Current
Voltage/mV
current/I
y = 6.6709x
y = 4.5918x
y = 2.6782x
0
1
2
3
4
5
6
7
8
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6

CALCULATION

For the line $y = 2.678x$(that is 36cm connecting cord)

$$slope = \frac{\Delta y}{\Delta x} = \frac{(3 - 1.6) \times 10^{-3} V}{(1.12 - 0.6) I} = 2.692 \times 10^{-3} \Omega$$

For the line $y = 4.591x$ (that is 61cm connecting cord)

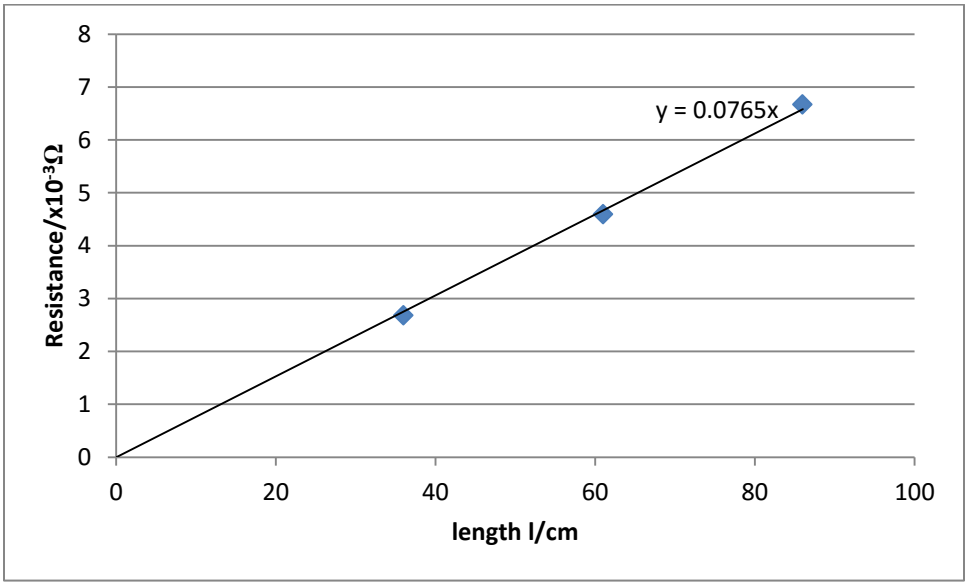
$$slope = \frac{\Delta y}{\Delta x} = \frac{(4.6 - 2.4) \times 10^{-3} V}{(1 - 0.52) I} = 4.583 \times 10^{-3} \Omega$$

For the line $y = 6.670$ (that is 86cm connecting cord)

$$slope = \frac{\Delta y}{\Delta x} = \frac{(4.8 - 2.4) \times 10^{-3} V}{0.72 - 0.36} = 6.667 \times 10^{-3} \Omega$$

DISCUSSION

The slopes of the graph, according to the formula in the theory ($\Delta V = IR$), represent the resistances of the three connecting wires used in the experiment. This means that using their current/voltage characteristics, the resistances of wires were determined. For the 36cm wire, the resistance of the wire was found to be 2.692x10$^{-3}\Omega$. For the 61cm wire, the resistance of the wire was found to be 4.583x10$^{-3}\Omega$. Lastly for the 86cm wire, the resistance of the wire was found to be 6.667x10$^{-3}\Omega$. A relationship between the length of the wire and the resistance can be seen. As the length of the wire increases, the resistance in the wire also increases provided the wires are made of the same material ($l \propto R$) where l is the length of the wire.



So as can be seen from this second graph, the slope is 0.076Ω/cm. This means that for every 1cm increase in the length of the wire, the resistance increases by 0.076Ω for these types of wires.

SOURCES OF ERROR

Due to the use of the amplifier, there might have been wrong conversion of the values read into the actual values.

PRECAUTIONS

1. Each time a different connecting wire was used the circuit was checked again for rightful connection.
2. The experiment was repeated twice for consistent values.

CONCLUSION

The resistance of the three wires of length 36cm, 61cm and 86cm are 2.678, 4.591 and 6.67 respectively.

REFERENCE

Walter C. MICHELS et al., PHYSICS: Principles and Application; Houghton Mifflin Company, Boston, 1977. Page 494-503.